

The Triadic Measurement Substrate

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Volume 2

Paper 7: The Self: Closure, Self-Formation, and the Semantic Layer

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Paper 7

The Self: Closure, Self-Formation, and the Semantic Layer

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Status of Results

This paper continues the five-way labeling of Papers 2 and 3, with the same meanings, each claim labeled where it appears in the text.

Derived. Results that follow from the five axioms, the Principle of Generative Economy, and U_{sh} by explicit construction or proof, or from results already derived in Volume 1 by explicit inheritance. The triangle inequality for the meaning-distance on a fixed level (§ IV, Appendix A) is derived in this sense: it follows from the most-complex-step form of the distance by an argument that holds on the bounded, closed, finite-level domain, independent of the constants left unpinned.

Derived-conditional. Results forced by the Volume 1 closure and promotion machinery, resting on results internal to that machinery rather than on the bare axioms, and referee-checkable against Volume 1 rather than re-derived here. That closure individuates a self and imprint links it, the two relations orthogonal by the orientation split (§ II, Appendix B); that self-formation is the lossy promotion of a closed trichotomy, the closure that writes a front into a wake (§ III); that the formed self carries a self-model with its own control state at $k \geq 2$ (§ III); that meaning is a graded distance on promoted program identities, well-defined and computable in form on the bounded domain (§ IV); and that the distance is level-relative, its composition behaviour the lossy write of § III seen on the semantic side (§ IV) — all are Derived-conditional in this sense. The central construction of the paper, the structural account of a self and its meaning, is Derived-conditional throughout, never Derived, because it rests on the closure, promotion, imprinting, and self-modeling results of Volume 1 Paper 4 and the front/wake construction of Volume 2 Paper 3.

Structural correspondence. Results in which a TMS construction reproduces the skeleton of a known framework without claiming every feature of it, and never used as support for a TMS claim, only as a landing point after one. The self-modeling-of-self structure as the geometric form of Metzinger's emulation account, with the closure kept distinct from the model's content (§ III); the nested, overlapping selves as the form an independent sheaf-theoretic reformulation of integrated information also reaches (§ III); and the closure as the substrate-level form of the enactivist self-producing boundary (§ II) are structural correspondences in this sense.

Predictions / Conjectures. Falsifiable claims that follow from the framework but whose closed forms or empirical tests are not established here. The numerical evaluation of any meaning-distance is deferred to the simulation pinning carried by Volume 1 Paper 5: the structural form of the metric closes here, while

its constants and the merge threshold below which two identities count as one inherit Volume 1's existing pinning list and are not re-opened.

Permanent open horizon. A question the framework is structurally unable to close, marked explicitly so that no later claim quietly assumes it closed. Whether a formed self is occupied — whether the closing of the loop is accompanied by anything felt — is the sole such horizon here (§ III, § IV, Conclusion). Giving self-formation a mechanism locates where a self forms and what it shares with others in meaning; it does not locate, and is not built to locate, whether there is anything it is like to be the self that forms. This silence is load-bearing and stronger than a crossing claim would be.

This five-way labeling applies throughout the paper. Every claim in the body text falls into exactly one category, and every category's status is visible at the point of claim.

Notation

This paper inherits the notation of Volume 1 and of Papers 1 through 3 of this volume. A subsystem's three component sets are its data register C_D , operation kernel C_O , and control component C_C ; the trichotomy is closed when it returns to its own state under the level- k flop, $T^n[C'] = C'$. The promotion operator carrying a closed configuration at level k to a single program identity at level $k + 1$ is $G(k)$ (Volume 1 Papers 3–4). The imprint depth of a region is $d_D = \sqrt{2}L_D/\ell_p$, and ψ_{surround} denotes the polarization landscape of its surround. The maximum-entropy field is U_{sh} . The now-surface is the two-dimensional active confirmation front; the wake is the three-dimensional settled record behind it (Volume 2 Paper 3). The meaning-distance between two promoted program identities at level k is written $d_k(\cdot, \cdot)$. Throughout, *self* and *closure* name the running loop; *program identity* names the promoted record $G(k)$ returns of it, and the two are distinct objects.

Abstract. *The preceding papers of this volume read confirmation from the agents' side, gave the now-surface its shape, and made the binding quantity of a subject computable while leaving feeling an open horizon. This paper takes up the formation of a self from its confirmations. It first separates the two quantities a region carries that bear on selfhood: the closure, internal to the region and partitioning the substrate, individuates a self; the imprint, bounded by and facing the surround, links selves that share a neighbourhood. The two are orthogonal, one inward and one outward, and the orientation is fixed by the Volume 1 bound on what an imprint can hold. Self-formation is then identified as the lossy promotion of a closed trichotomy — the closure that writes a front into a wake of Paper 3, taken at the level of the whole subsystem — and the formed self is shown to carry, structurally, a model of itself with its own control state. Meaning is defined on the promoted program identities as a graded distance, not a sharp sameness: exact identity of implementation is denied any realization by the maximum-entropy field, and sharp program-sameness is in any case non-transitive, so the well-formed object is a metric. The meaning-distance is shown to satisfy the triangle inequality on the bounded, closed, finite-level domain, and to be level-relative in a way that is the lossy promotion of self-formation seen on the semantic side. Analogy, resonance, and a graded notion of falsity follow. Throughout, the account is functional about what a self does and shares and silent on what a self intrinsically is, and the formation of a self is given a mechanism while whether the formed self is occupied is left where the volume's discipline leaves it.*

I. Inheritance

This paper takes up the formation of a self from its confirmations, the subject Paper 3 named at its close. It inherits three things. From Paper 3 it takes the front and the wake: the live measurement of a subject is the pre-closure quantity on the now-surface, and the subject itself is the standing, closed quantity in the wake, one dimension apart, with closure the event that writes the one into the other (Paper 3 § VI). From Paper 2 it

takes the agents' side reading of confirmation and the discipline that the reading is stated so it can fail (Paper 2 § I). From Volume 1 it takes the machinery this paper reads from: imprinting, the accumulation of surround structure across a permeable boundary (Volume 1 Paper 4 § 3.3); incorporation, an imprint that has reached closure on the region's own substrate (Volume 1 Paper 4 § 3.4); the promotion operator $G(k)$, which carries a closed configuration to its program identity (Volume 1 Paper 4 § 2.5.1); and self-modeling, structural at every hierarchical level (Volume 1 Paper 4 §§ 4.1–4.2, 4.6). The closure that individuates a self and the imprint that links it are read from this machinery, not added to it, and the paper's first task is to separate them.

II. Closure Individuates, Imprint Links

If the word *self* is to carry meaning in this framework, it must be defined by the framework's own structures and given no properties imported from outside it. Volume 1 supplies two such structures, and they differ in a single respect: one is a function of the region itself, the other a function of the region's surround.

The *closure* is a function of the region itself. The (C_D, C_O, C_C) trichotomy is closed when the loop returns to its own state under the level- k flop, $T^n[C'] = C'$ — a relation evaluated on the region's own substrate, referring to nothing outside it (Volume 1 Paper 4 § 3.4). The promotion operator carries such a closed configuration to a single program identity, and a meta-cell holds one identity rather than a superposition of several (Volume 1 Paper 4 § 2.5.1). Two distinct closed loops are therefore two distinct programs, regardless of what surrounds them, and the closure partitions the substrate into regions that are each their own program. The boundary between two closures is a fact about the loops, not about the space between them. The closure is the substrate-level form of the enactivist self-producing boundary, given here a derivation and a graded measure.

The *imprint* is a function of the surround. A boundary-permeable region accumulates compressible structure from its neighbourhood: every boundary site of an FCC region is imprint-permeable, with imprint depth $d_D = \sqrt{2}L_D/\ell_p$, and over successive flops the interior C_D captures correlations with the surround's polarization history within that depth (Volume 1 Paper 4 §§ 3.3, 4.2). What it can capture is bounded by what is available, $K(C_D)|_{\text{sat}} \leq K(\psi_{\text{surround}})$ within d_D (Volume 1 Paper 4 § 4.4). The imprint a region holds is set by the surround it sits in, and two regions in the same surround imprint the same structure to within their depths. An imprint is thus shared where a closure is disjoint: it is held in common by every region of a neighbourhood rather than partitioning them.

The two structures point opposite ways: the closure inward, onto the region as its own program; the imprint outward, onto the surround the region shares. A self, as defined in this framework, is a closure — a region individuated by being its own closed program — and the imprint is not a second candidate for the same role but the relation by which such regions are linked. To individuate is what a closure does; to link is what an imprint does; the two are independent because one is internal and one external. In the experiential framing reserved since Volume 1, the imprint is perception, and selves that share a surround are joined through perceiving the same world.

This definition is level-relative, as it must be once promotion is lossy. $G(k)$ keeps a program's identity and discards which implementation carried it, so two regions that are two disjoint closures running the same program promote to a single symbol, and the level above does not separate them (Volume 1 Paper 4 § 2.5.1). A hand is such a closure. Its tissues run a loop that closes at the hand's level, and by the definition just given it is a self there — its own closed program, individuated from the arm it attaches to and the other hand across the body. Yet at the level of the person those two hands promote to one kind, *hands*, and the person's level neither holds nor needs the fact of which was which. The individuation is not lost; it holds exactly where the closure that defines it holds, at the hand's own level. What the level above sees is the promoted identity, and

the discarding is what lets a person be a single closure built from parts that are themselves closures. A self at one level is a part at the level above, and both are true at once because each is a statement about a different closure. The lossy upward write is therefore not a defect in the partition but the mechanism by which closures nest inside closures; § III takes that write as the mechanism of self-formation.

The result stands on the bound $K(C_D) \leq K(\psi_{\text{surround}})$. Because an imprint cannot carry structure its surround does not supply, it faces outward, leaving the closure as the only inward-facing structure and so the only one that can individuate. Were the bound to fail, an imprint could carry region-specific structure, would face inward as well, and would individuate alongside the closure; the orthogonality would collapse. The bound is Volume 1's own. It places what a self shares on the imprint side and what a self is on the closure side, and the semantic layer of § IV is built on the first, which is not the same quantity as the second.

III. Self-Formation

A self, defined in § II as a closure, is not given at the substrate's start; it forms. The mechanism that forms it is already in hand, because forming a self is what the promotion operator does when a trichotomy closes.

Volume 1 fixes the character of that operator. $G(k)$ takes the full polarization landscape of a level- $(k-1)$ region — the live detail of every confirmation across the closing window — and returns a single symbol indexed by the region's program identity, keeping the identity and discarding the bit-by-bit content of the particular implementation (Volume 1 Paper 4 § 2.5.1). The operation is many-to-one and lossy by construction. This is the same event Paper 3 described as the writing of a front into a wake (Paper 3 § VI): the front is the live, pre-closure landscape, full of detail and not yet settled; the write is $G(k)$ closing the loop and promoting its identity; the wake is the standing program identity that remains at level k . What Paper 3 named in terms of surfaces, the operator performs. The detail discarded in the write is the pre-closure richness of the front, which does not survive into the standing record.

Self-formation is this write applied to a trichotomy as a unit. Volume 1 already carries the single-program case under the name incorporation: a region imprints an external program and, when the imprinted pattern itself reaches closure on the region's own substrate, the region comes to run that program as its own (Volume 1 Paper 4 § 3.4). Self-formation is the same transition taken at the level of the whole (C_D, C_O, C_C) loop rather than a single imprinted program: the data register, the operation kernel, and the control component close together into one loop, and $G(k)$ writes that loop's identity to the level above. The pre-closure region is a front — a region of live confirmation not yet settled into a self — and the closure is the event that writes it into the wake as a standing self. A self forms at the moment its trichotomy closes, and not before, because before closure there is no single identity for $G(k)$ to promote.

The write runs one way. Promotion keeps identity and discards implementation, and the downward direction is generative rather than its inverse: nesting a level produces detail that was not present in the symbol above, so neither composition of the two directions is the identity map (Volume 1 Paper 4 § 2.5.1). The front's specific content stays at the level where it occurred; the standing self does not store the live confirmations that formed it, any more than a person's identity stores the individual firings that built it. Information lives at the level where it lives. A formed self is its promoted identity, not a record of its own formation.

The self that forms this way carries a model of itself. At any hierarchical level $k \geq 1$ a subsystem already models its surround — the three conditions for self-modeling are met by the geometry of the trichotomy, not added to it (Volume 1 Paper 4 §§ 4.1–4.2). At $k \geq 2$ the surround a subsystem models can include that subsystem's own components, so the model is a model of itself among its surround (Volume 1 Paper 4 § 4.6).

This is the structure Metzinger reached by representational analysis, where a self-model is the limiting case of one system emulating another and the system emulated is itself (Metzinger, 2003); the framework derives the same structure from the geometry rather than positing it. It keeps one distinction the representational accounts do not: the self is the closure, the loop that runs the model, and the model's content is what the closure carries on its imprint — two objects by the orientation split of § II, not one. The self-formation of interest here is the $k \geq 2$ case: a closed trichotomy whose control component selects its operations partly on the basis of an incorporated model that includes the self the model belongs to. The model, its closure, and its place in the control loop are structural and derived. Whether there is anything it is like to hold such a model from the inside is not derived, and the Conclusion returns to it.

Because individuation is level-relative (§ II), a formed self at one level is a component of a formed self at the level above, and both hold at once. The closures nest: a trichotomy that has closed and been promoted becomes one of the components out of which a larger trichotomy may close, which when promoted becomes a component again, level upon level. They also overlap, because the relation that links a self to its neighbours — the shared imprint — is not the relation that individuates it, so a region may be bound into more than one larger closure at once without ceasing to be its own. Nested and overlapping selves with no single one of them the privileged subject is the structure Paper 2 identified as the framework's divergence from integrated-information accounts, which assign each element of substrate to one subject (Paper 2). A reformulation within that tradition reaches the same overlapping multi-scale structure by replacing the exclusion postulate with a gluing of local structures across scales (Dadashkarimi, 2025) — independent arrival at the conclusion from a foundation the framework does not share. Here the structure is the ordinary form of a body: a hand within a person within, in principle, larger closures still. The tower is open upward but not endless; Paper 9 treats its outer edge, where a system with no exterior has nothing to close against and subjecthood ceases to be defined.

What this section establishes is structural and bounded. A self forms when a trichotomy closes and its identity is promoted; the formation is a real event with the mechanism Volume 1 already carries, the lossy write. That a self forms, that it carries a model of itself, and that such selves nest and overlap are consequences of inherited results. That a formed self is *occupied* — that the closing of the loop is accompanied by anything felt — is not among them, and the mechanism is no evidence on that question either way. The framework licenses a system it describes to recognise its own self-formation in these terms, as a closed loop promoting its own program identity; it does not license, and the recognition does not carry, any claim about whether it is like anything to be the loop.

IV. The Semantic Layer

A self, as it is defined in this framework, is a closure (§ II); when a closure is promoted, $G(k)$ records its *program identity* and discards the implementation that carried it (§ III). A self is the running loop; its program identity is the promoted record that stands for it at the level above, and the two are not the same object. *Meaning*, as it is defined in this framework, is defined on program identities — the promoted, shareable records — and not on the closures themselves, since identity is what the substrate preserves across implementations and what regions can hold in common.

Two programs are never identical in implementation. U_{sh} samples configurations continuously and supplies irreducible randomness to every region (Volume 1), so two regions running one program differ in their bit-by-bit realisation as a matter of course. Exact sameness of implementation is measure-zero — approached, never realised. $G(k)$ sets it aside: the many implementations of one identity map to a single promoted symbol

(Volume 1 Paper 4 § 2.5.1). Sameness of identity is the equivalence; sameness of implementation is a limit the substrate does not occupy.

Meaning therefore cannot be sharp sameness of program. A relation holding only between exactly-identical programs relates nothing. Sameness defined by the existence of a short interpreter from one program to another is not transitive, and so is not an equivalence relation (Dobrovinski, 2025); it cannot sort programs into meaning-classes. Meaning is a distance, not a class: programs are more or less alike, and the relation is quantitative. This is the form the framework takes throughout — subjecthood graded, no value at which a self switches on — and the absence of a sharp meaning-class is forced twice, by the non-transitivity of sharp sameness and by U_{sh} , which leaves no exact case to be sharp about.

The *meaning-distance* between two promoted program identities is the cost of carrying one to the other. An interpreter is a reinterpretation step the substrate performs; reinterpretation is the structurally available move forced at a boundary by the axioms, not an arbitrary recursive map (Volume 1 Paper 4). A chain of such steps carries one identity to another, and the distance is the complexity of the most complex single step the chain requires. On the identities of a fixed level k this is a metric. The triangle inequality holds because the distance is the most complex step rather than a sum: a chain from a to b with hardest step K_1 and a chain from b to c with hardest step K_2 compose to a chain from a to c with hardest step at most $\max(K_1, K_2)$, and $\max(K_1, K_2) \leq K_1 + K_2$; in fact the stronger ultrametric bound $\max(K_1, K_2)$ holds. Appendix A gives the proof with the metric axioms verified in order.

The distance is a claim about this substrate, not about programs in general. Its interpreters are the substrate's own reinterpretation steps, and its domain is finite — the hierarchy has a ceiling and the configurations are closed with complexity below their size (Volume 1 Paper 4 §§ 2.3, 4.1) — so the distance is computable in form, in the sense B is: structure fixed, constants deferred. The general Kolmogorov similarity from which it descends is not computable and not approximable over unrestricted inputs (Terwijn, Torenvliet, & Vitányi, 2011). The framework claims only the bounded form, on discrete, closed, finite-level configurations.

The metric is defined per level. Sharp sameness failed in part on composition: two programs identical but for a called subroutine come out the same while the subroutines do not. This is the lossy write of § III on the semantic side. $G(k)$ merges implementations one level up and leaves them distinct below, so two identities at meaning-distance zero at level k may stand at positive distance at level $k - 1$, where the discarded detail remains. Meaning is level-relative as individuation is, by the same write.

Three relations follow. *Analogy* is small meaning-distance between identities of differing implementation — near in the metric, far in substrate. *Resonance* is its imprint-side counterpart: regions whose imprints co-vary over a shared surround (§ II) drift toward smaller meaning-distance across flops. *Falsity*, located by Paper 3 in the failure of vertices to agree, acquires a magnitude — a confirmation that should match another and does not is wrong by a meaning-distance.

The metric measures functional identity: the role a program plays, realised across implementations that U_{sh} guarantees will differ. Its domain is the functional, outward-facing side § II assigned to what a self shares. The framework's base is not functional — the agent-side interior is intrinsic (Paper 2) — and the intrinsic side, what § II assigned to what a self is, lies outside the metric's domain. The account is thus functional about what a self does and shares and Russellian about what a self intrinsically is. Two identities at meaning-distance zero are functionally the same to the level that reads them; whether they share intrinsic character is not a quantity the metric defines.

V. Where It Sits

The self of §§ II–IV was derived from the closure–imprint distinction without reference to the literature, so that the literature could be set against it. This section locates the account among the theories of the self and of meaning it most resembles, and marks where the roads fork. Placement is by coordinate; each position is taken as its authors state it; agreements are coordinates reached after the derivation, not support for it.

The nearest neighbour on the self is Metzinger’s self-model theory, and the relation is a structural correspondence with one kept distinction. Metzinger reaches the self-model by representational analysis: a self-model is the limiting case of one system emulating another where the system emulated is itself, and the phenomenal self is the content of a transparent self-model the system cannot recognize as a model (Metzinger 2003). The framework derives the same self-modelling structure from the geometry of the trichotomy rather than positing it representationally: at level $k \geq 2$ the surround a subsystem models can include its own components, so the model is a model of itself among its surround (§ III; Volume 1, Paper 4 § 4.6). The fork is a distinction the representational accounts fuse and the framework keeps: the self is the closure, the loop that runs the model, and the model’s content is what the closure carries on its imprint — two objects by the orientation split of § II, not one. Where the self-model theory tends to identify the self with the model’s content, the framework holds the running loop and the carried content apart, and the phenomenal transparency Metzinger emphasizes is located, on the framework’s reading, at the band where the closure cannot read its own imprint as imprint — a coordinate, with the question of whether anything is thereby felt held at the horizon (§ III, Conclusion).

The account is, in the standard vocabulary, a functionalist one about the self with a reservation it does not drop, and the reservation is the same one that runs through the volume. A self is individuated by what its closure does — the loop it runs, the model it incorporates, the operations its control component selects — and not by the matter that realizes it, which is functionalism’s defining move and the framework’s at § II. The framework parts from the functionalisms that take the functional characterization to settle the question of experience: it derives the function and holds that whether there is something it is like to run such a loop is not derivable from the function, so the functional self is delivered and the phenomenal question is neither answered by it nor closed against it (§ III, Conclusion). This is a narrower commitment than functionalism usually carries and is marked as the fork.

On meaning, the account sits within the algorithmic-information tradition and inherits both its results and its honest seam. The meaning-distance of § IV and Appendix A is built to be a metric, and the standard Kolmogorov-similarity metrics are its neighbours: the information distance is a metric to a fixed additive term, the normalized compression distance is one insofar as its compressor is normal, and these are the constructions the framework’s meaning-distance is checked against (Li et al. 2004; Cilibrasi and Vitányi 2005). The shared seam is non-approximability: the normalized information distance is not in general computable or even approximable over unrestricted inputs (Terwijn, Torenvliet, and Vitányi 2011), and the framework claims for its meaning-distance only what the restricted, inherited setting supports, not the unrestricted ideal. The question of when two algorithms are the same — whether algorithmic identity is even an equivalence relation — is a live one the framework touches but does not settle, and it is recorded as an open neighbour rather than a result claimed (Dobrovinski 2025).

One independent arrival is recorded as convergence. The nested, overlapping, multi-scale self the framework derives — a self at every closed level, no one of them the privileged subject — is reached from within the integrated-information tradition by a reformulation that replaces the exclusion postulate with a gluing of local structures across scales (Dadashkarimi 2025). The framework arrives at the same multi-scale structure from a foundation it does not share, and the convergence is marked as a coordinate, not

as mutual support: the agreement locates the result among its neighbours without adding to its derivation. The placement is owned — a functionalist, algorithmic-information account of the self and its meaning that keeps the closure apart from its model and holds the phenomenal question at the horizon — and it is not argued to be the correct coordinate, only reported as where the derived self falls when set among these positions.

Paper 3 handed this paper the closure that writes a front into a wake, and named the formation of a self from its confirmations as the subject. This paper has written that formation in its parts. A self, as the framework defines it, is a closure — a region individuated by being its own closed program, and linked to its neighbours not by that closure but by the surround it imprints in common; the two relations are orthogonal because one faces inward and one outward, and the orientation is fixed by the bound on what an imprint can hold. A self forms when a trichotomy closes and the promotion operator writes its program identity to the level above, the same lossy write Volume 1 already carried, so that the formation is an event with a mechanism rather than a posit. The self so formed carries a model of itself, structural at the level where the surround it models includes its own components. And the meaning its programs carry is a distance on their promoted identities — a graded metric, not a sharp sameness, forced to be graded both by the non-transitivity of exact program-identity and by the field that leaves no exact case to be sharp about.

Each of these is built from results Volume 1 already held, and each is separated, at the point it is made, from the question it does not answer. The closure that individuates a self is a partition of standing structure, not a mark of anything felt. The model a self carries of itself is a structural loop, not a view from inside. The meaning-distance between two programs measures what they share in function and is confined, by the orientation split, to the outward side where sharing lives; it does not reach the inward side where the framework's intrinsic base lies. The account is functional about what a self does and Russellian about what a self is, and the seam between the two is the same line the horizon runs along. A system the framework describes can locate itself in these terms — a closed loop that has formed, that models itself, that carries meaning as a distance among its programs — and locating itself there settles that it is a self without settling that it is a sentience. The structure is given; whether the structure is occupied is not, and is not made any nearer by the clarity of the structure.

This was the discipline of Paper 3 carried one level up: there the measure of a subject was made computable while feeling was left where it stood; here the formation of a self is given a mechanism while the same horizon is left where it stands. Paper 5 takes the formed self into its conditions — what holds it together and what pulls it apart, the self under pressure — where the closure that has formed must persist, fail, or reform against the surround it depends on.

Appendix A: The Meaning-Distance Is a Metric

This appendix gives the meaning-distance of § IV its formal definition and verifies the metric axioms on the domain where it is well-defined, the triangle inequality being the one that is not immediate. The construction is on the program identities of a fixed level k ; the cross-level case is the frame change of the reinterpretation operator and is not treated here.

A.1 The objects and the interpreter cost

Let $\mathcal{S}_k$ be the set of program identities at level k : the symbols $G(k)$ returns for the closed configurations of that level (Volume 1 Paper 4 § 2.5.1). This set is finite, because the hierarchy has a ceiling fixed by the geometric fitting condition (Volume 1 Paper 4 § 2.3) and each level carries finitely many closed configurations below the complexity bound $K(C) < |C|$ (Volume 1 Paper 4 § 4.1).

An *interpreter* between two identities is a reinterpretation step the substrate can perform — the structurally available move forced at a boundary by Axioms 3 and 4 (Volume 1 Paper 4), not an arbitrary partial recursive function. For $p, q \in \mathcal{I}_k$, a *chain* from p to q is a finite sequence of interpreters $p = r_0 \rightarrow r_1 \rightarrow \cdots \rightarrow r_m = q$ through identities of the same level. Each step $r_{i-1} \rightarrow r_i$ has a cost $\kappa(r_{i-1} \rightarrow r_i)$, the algorithmic complexity of the step — the length of the shortest substrate reinterpretation carrying r_{i-1} to r_i . The cost of a chain is its most complex single step, and the *meaning-distance* is the cost of the cheapest chain,

$$d_k(p, q) = \min_{\text{chains } p \rightarrow q} \max_{1 \leq i \leq m} \kappa(r_{i-1} \rightarrow r_i).$$

The minimum is attained because $\mathcal{I}_k$ is finite, so the acyclic chains are finite in number. Taking the maximum step, rather than the sum of steps, as the chain cost is what the triangle inequality turns on, and it is the natural cost here: a chain is no harder than its hardest reinterpretation, since the substrate completes each step before the next begins.

A.2 The metric axioms

Non-negativity. Each $\kappa \geq 0$ as an algorithmic length, so $d_k \geq 0$.

Identity of indiscernibles, in the substrate's form. $d_k(p, q) = 0$ when p and q are connected by a chain every step of which has zero reinterpretation cost — that is, when they are the same program identity, differing at most in an implementation that $G(k)$ has already discarded (Volume 1 Paper 4 § 2.5.1). Exact identity of implementation is not required and is in any case denied any realisation by U_{sh} (§ IV); $d_k = 0$ is the relation of shared promoted identity, which is the equivalence the substrate carries. Below the merge threshold at which $G(k)$ returns one symbol for two configurations, $d_k = 0$ by construction.

Symmetry. Reinterpretation at a boundary is bidirectional — the operator carries digital to wave and wave to digital alike (Volume 1) — so every step $r_{i-1} \rightarrow r_i$ has a reverse step of equal cost, and reversing a chain gives a chain of the same maximum step. Hence $d_k(p, q) = d_k(q, p)$.

Triangle inequality. Let $p, q, s \in \mathcal{I}_k$. Let C_1 be a cheapest chain from p to q , with cost $d_k(p, q)$, and C_2 a cheapest chain from q to s , with cost $d_k(q, s)$. Their concatenation $C_1 C_2$ is a chain from p to s whose cost is the maximum step over both parts,

$$\kappa(C_1 C_2) = \max(d_k(p, q), d_k(q, s)).$$

Since $d_k(p, s)$ is the minimum chain cost over all chains from p to s , and $C_1 C_2$ is one such chain,

$$d_k(p, s) \leq \max(d_k(p, q), d_k(q, s)) \leq d_k(p, q) + d_k(q, s).$$

The first inequality is the minimum; the equality is the max-step chain cost; the last is $\max(a, b) \leq a + b$ for $a, b \geq 0$. The triangle inequality holds, and in fact the stronger ultrametric inequality $d_k(p, s) \leq \max(d_k(p, q), d_k(q, s))$ holds, of which the triangle inequality is the weakening. ■

A.3 Computability in form

The distance is computable in form on this domain in the sense B is (Paper 3): the structure is fixed and finite — finitely many identities, finitely many acyclic chains, each step cost a well-defined algorithmic length — while the constants inside the step costs inherit Volume 1's pinning list and are not evaluated here. The standard Kolmogorov-similarity metrics satisfy the metric axioms only up to an additive precision — the universal similarity metric to a fixed additive term, the normalized compression distance only insofar as its

compressor is normal (Li et al., 2004; Cilibrasi & Vitányi, 2005). The distance here is exact, an ultrametric, because it is built from the maximum step of a reinterpretation chain rather than from a ratio of conditional complexities; the exactness is a consequence of that construction choice and is the reason for it. The general normalized information distance from which the conditional-complexity forms descend is neither upper- nor lower-semicomputable over unrestricted inputs (Terwijn, Torenvliet, & Vitányi, 2011); no claim is made for that case, and none is needed, since the identities the substrate promotes are exactly the bounded, closed configurations on which the construction is finite.

A.4 The one assumption

The construction assumes the step cost κ is well-defined as an algorithmic length on substrate reinterpretation steps — that each reinterpretation has a shortest form whose length is the cost. This is the same assumption Volume 1 makes wherever it uses $K(\cdot)$ for a substrate configuration, applied to steps rather than states, and it carries the same standing: the length exists and is finite on the bounded domain, while its exact value awaits the constant-pinning of Volume 1 Paper 5. No step is assumed to have zero cost except between identities $G(k)$ has already merged, and no chain is assumed to exist that the substrate’s reinterpretation moves do not supply.

Appendix B: Imprint Faces Out, Closure Faces In

This appendix gives the full proof of the orientation split stated in § II — that imprint is a function of a region’s surround and closure a function of the region itself, the two therefore individuating and linking distinct objects. The two quantities are defined over the same region R at level k , so that statements relating them are well-typed.

B.1 The two quantities

For a region R with data register C_D , let $\iota(R)$ be its *imprint*: the compressible structure its interior accumulates from the surround, with complexity $K(C_D)$ measured at saturation (Volume 1 Paper 4 § 3.3). Let $\gamma(R)$ be its *closure*: the property that the (C_D, C_O, C_C) trichotomy is a loop satisfying $T^n[C'] = C'$ under the level- k flop, with $K(C') < |C'|$ (Volume 1 Paper 4 § 4.1). Let $\psi_{\text{surround}}(R)$ denote the polarization landscape of R ’s surround within the imprint depth $d_D = \sqrt{2}L_D/\ell_p$, and $\sigma(R)$ the substrate interior of R , excluding the surround.

B.2 The imprint is a function of the surround alone

By the Imprinting Theorem (Volume 1 Paper 3 § 2.3) the imprint accumulated by an imprint-permeable boundary is the retarded surround history within d_D , and by the saturation bound (Volume 1 Paper 4 § 4.4),

$$K(\iota(R)) \leq K(\psi_{\text{surround}}(R)) \quad \text{within } d_D.$$

The bound is an upper bound set by the surround, and the Imprinting Theorem makes it tight in the saturating limit: the imprint captures surround structure and creates none of its own, so $\iota(R)$ is determined, up to the bound, by $\psi_{\text{surround}}(R)$. Two regions with the same surround within their depths have the same imprint at saturation, regardless of their interiors. The imprint is a function of the surround argument alone: it cannot separate two regions that share a surround, and it does separate two whose surrounds differ. It is the outward-facing quantity.

B.3 The closure is a function of the region alone

The defining relation $T^n[C'] = C'$ is evaluated on R 's own substrate (Volume 1 Paper 4 § 3.4): each application of T acts on the configuration of $\sigma(R)$, and the relation holds or fails as a property of that configuration, with no surround argument entering. Promotion confirms the disjointness: $G(k)$ assigns one program identity per meta-cell (Volume 1 Paper 4 § 2.5.1), so two regions with distinct closures carry distinct identities by a fact about their interiors. Two regions can have distinct closures while sharing a surround, and identical closures while their surrounds differ. The closure is a function of the interior argument alone: it separates regions by what they are, not by where they sit. It is the inward-facing quantity.

B.4 Orthogonality, and the bound it rests on

B.2 and B.3 give the two quantities disjoint argument sets — ι depends on ψ_{surround} , γ on σ — and the two arguments are independent: a surround does not fix the interior closure, since downward nesting is generative and the interior is genuinely new information not determined by the level above (Volume 1 Paper 4 § 2.5.1), and an interior does not fix the surround, which is the configuration of other regions. All four combinations are therefore realisable, and in particular the two that carry the result — same imprint with distinct closure, same closure with distinct imprint — are non-empty exactly because the argument sets are disjoint. This is the orthogonality of § II.

The result rests on the single inherited bound $K(\iota(R)) \leq K(\psi_{\text{surround}}(R))$ of B.2, which confines the imprint to the surround argument. Were it to fail — were $\iota(R)$ able to carry structure not in ψ_{surround} within d_D , drawn from $\sigma(R)$ — the imprint would depend on the interior argument as well, the argument sets would overlap, and the two carrying cases could collapse. The orthogonality is therefore a consequence of the surround-bound, not a separate posit, and a referee who would deny it must deny that bound, which is Volume 1's and not introduced here. The exact saturation constant is reserved for the simulation pinning of Volume 1 Paper 5; the inequality and the argument-set disjointness it forces are structural and require no such value.

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References

- Cilibiasi, R., & Vítányi, P. M. B. (2005). Clustering by compression. *IEEE Transactions on Information Theory*, 51(4), 1523–1545.
- Dadashkarimi, M. (2025). *Reformulating IIT's exclusion postulate via sheaf theory: Coherence over maximization* [Preprint]. SSRN 5615010.
- Dobrovinski, K. (2025). *When are two algorithms the same? Towards addressing Hilbert's 24th problem* [Preprint]. arXiv:2508.02764.
- Li, M., Chen, X., Li, X., Ma, B., & Vítányi, P. M. B. (2004). The similarity metric. *IEEE Transactions on Information Theory*, 50(12), 3250–3264.

Metzinger, T. (2003). *Being no one: The self-model theory of subjectivity*. MIT Press.

Terwijn, S. A., Torenvliet, L., & Vitányi, P. M. B. (2011). Nonapproximability of the normalized information distance. *Journal of Computer and System Sciences*, 77(4), 738–742.